VERSION 1.1

2026

Project: Azure Cost Visibility Dashboard

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1. YANNICK HAMELRYCK

For a project I wanted to do something that would be beneficial in the real world.

This project focuses on creating an Azure Cost Visibility Dashboard to improve visibility into Azure spending. The goal is to build a practical solution that tracks cloud costs, shows where money is being spent, and helps prevent unexpected charges. So there won't be any unexpected invoices at the end of the month.

2. PROJECT GOAL

The goal of this project is to use Azure Cost Management, Azure Monitor, Logic Apps, and Azure Workbooks to create a dashboard and alerting system. The solution should make Azure costs easier to understand and notify stakeholders when spending reaches predefined thresholds.

3. BUSINESS PROBLEM

Many organizations move to the cloud expecting lower costs, but often struggle to understand their monthly Azure bill. Without clear visibility, it becomes difficult to know which services generate the highest costs or when spending is increasing unexpectedly. This project addresses that problem by creating a system that makes Azure spending easier to monitor, analyze, and explain.

4. AZURE SERVICES USED

I'll be using the free Azure account so some services are limited.

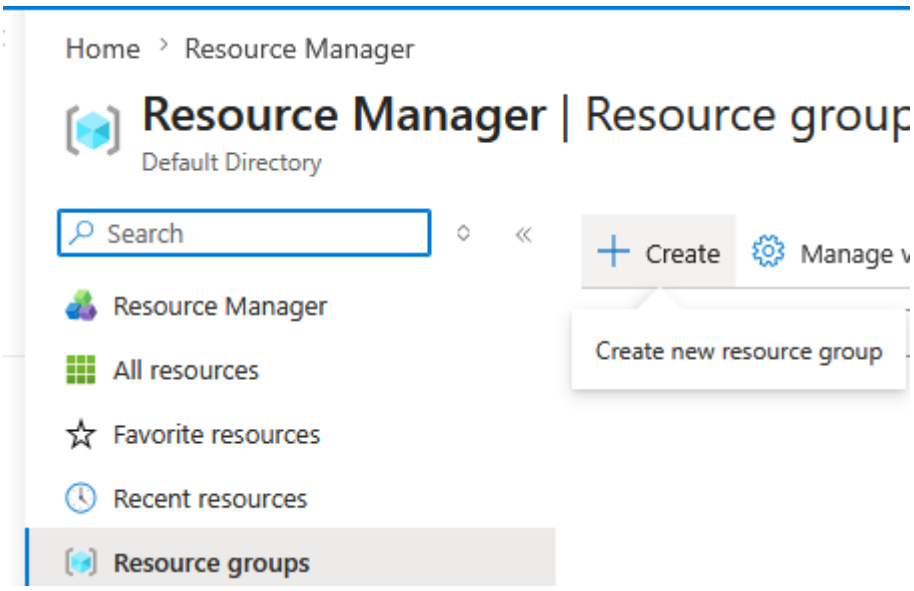
If I think about cost management there are a few Azure services that comes to mind:
Azure Cost management – Tracks Azure spending and shows costs breakdowns
Azure Monitor – Collects metrics and logs from Azure resources
Azure workbook – Create dashboards
Logic Apps – Automate actions (sending an alert email)
And Azure function – Runs small code on demand (process cost data or calculate daily spending)

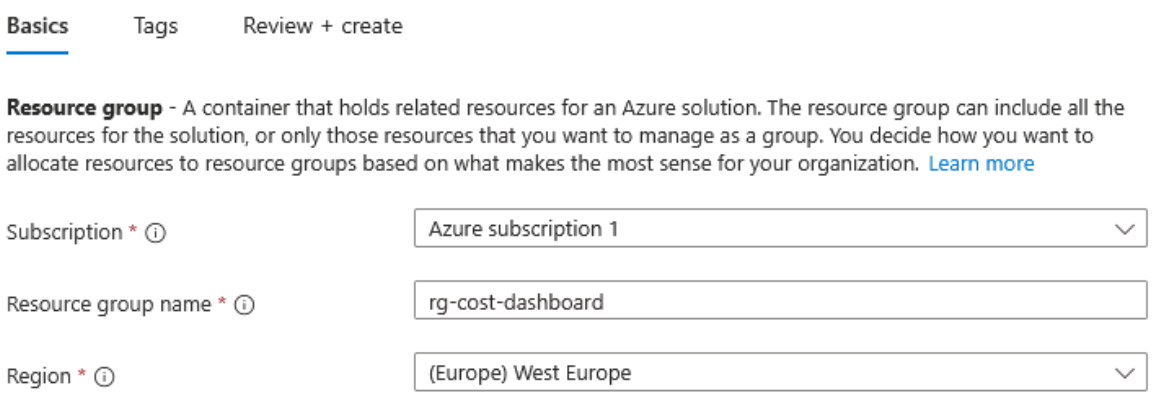
5. PLANNED WORKFLOW

1. Track Azure Spending using Azure Cost management
2. Group and interpret costs in a business readable way
3. Create alerts for selected spending thresholds
4. Trigger a Logic App to send notification
5. Visualize spending in a dashboard

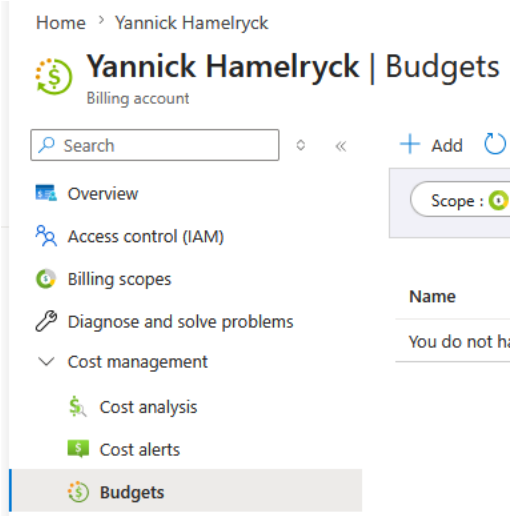
6. CONFIGURATION

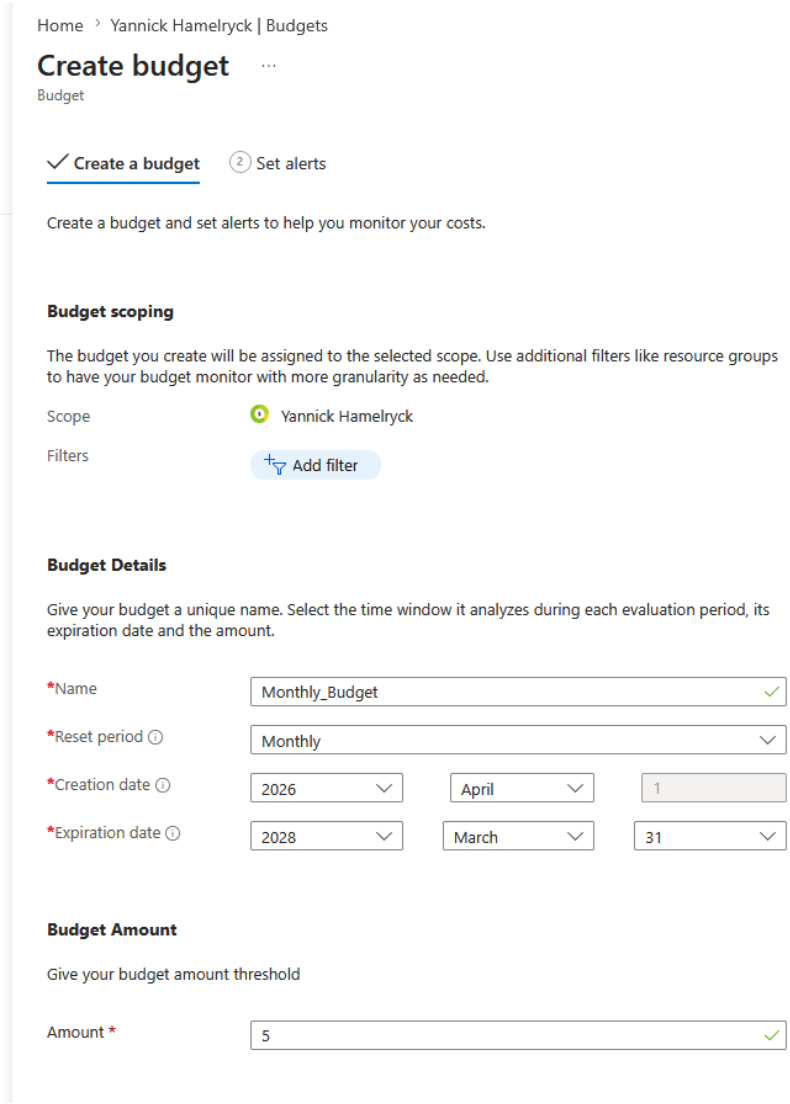
A separate resource group was created for this project so that all resources remain organized and easy to manage. I named the resource group rg-cost-dashboard and deployed it in West Europe. Using a dedicated resource group also makes it easier to monitor costs and remove all project resources afterwards.





The next step was creating a monthly budget in Azure Cost Management. I configured a small budget of $5 because this project uses a free Azure account and only needs enough spend to test the alerting functionality.

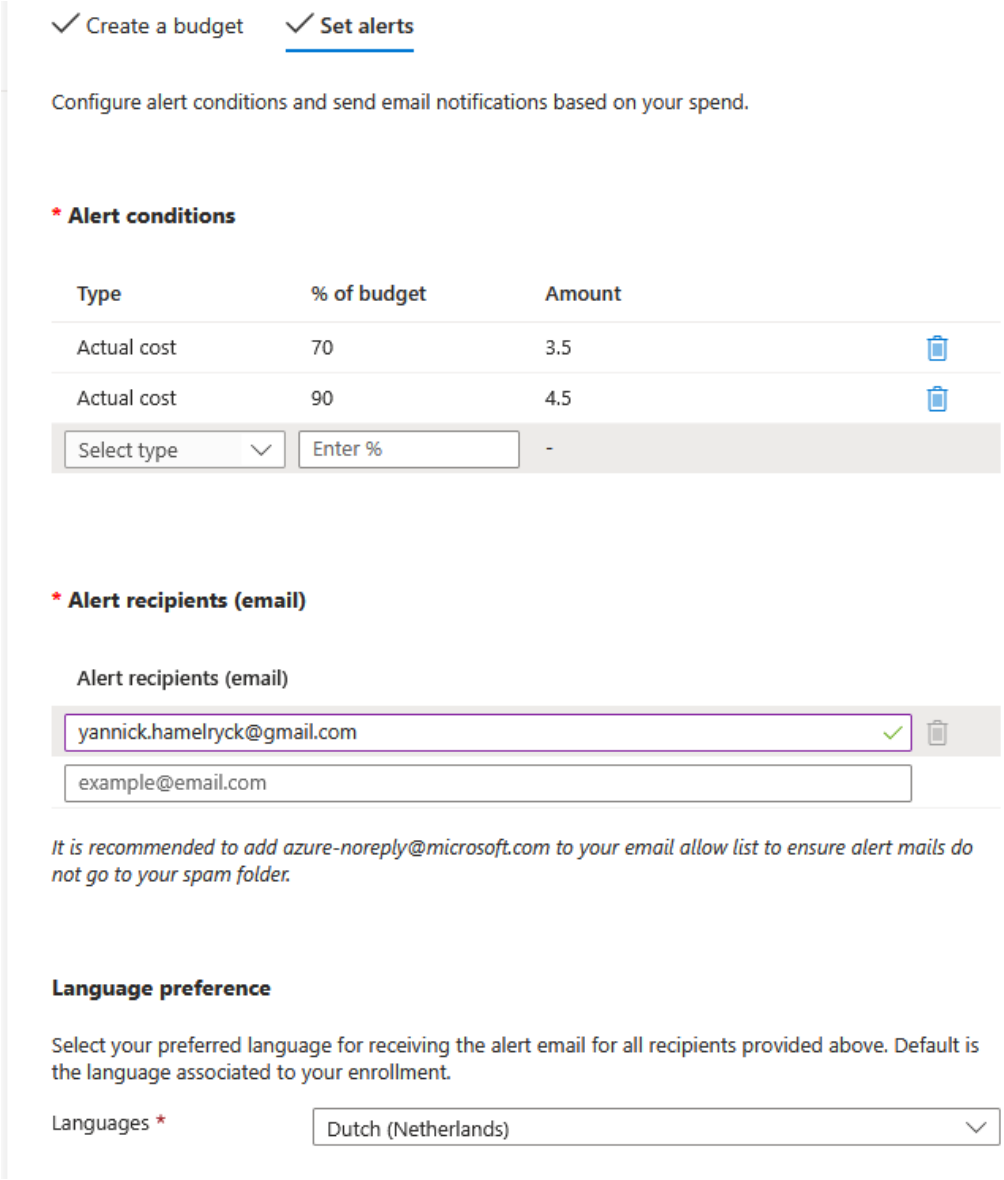




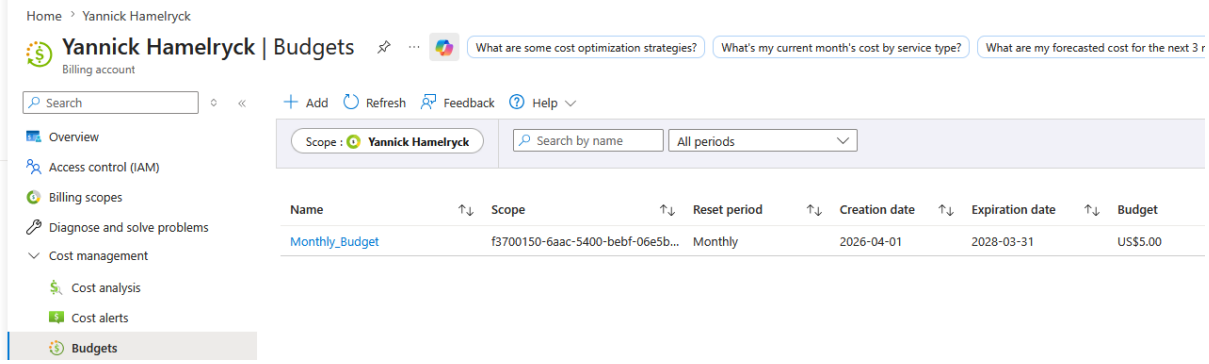
I configured two alert thresholds:

- 70% of the budget ($3.50)
- 90% of the budget ($4.50)

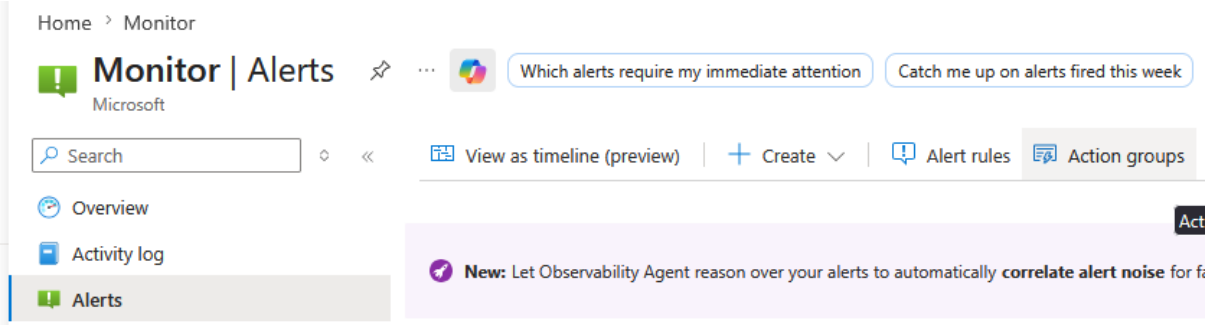
These thresholds make it possible to receive an early warning before the budget is completely exhausted.

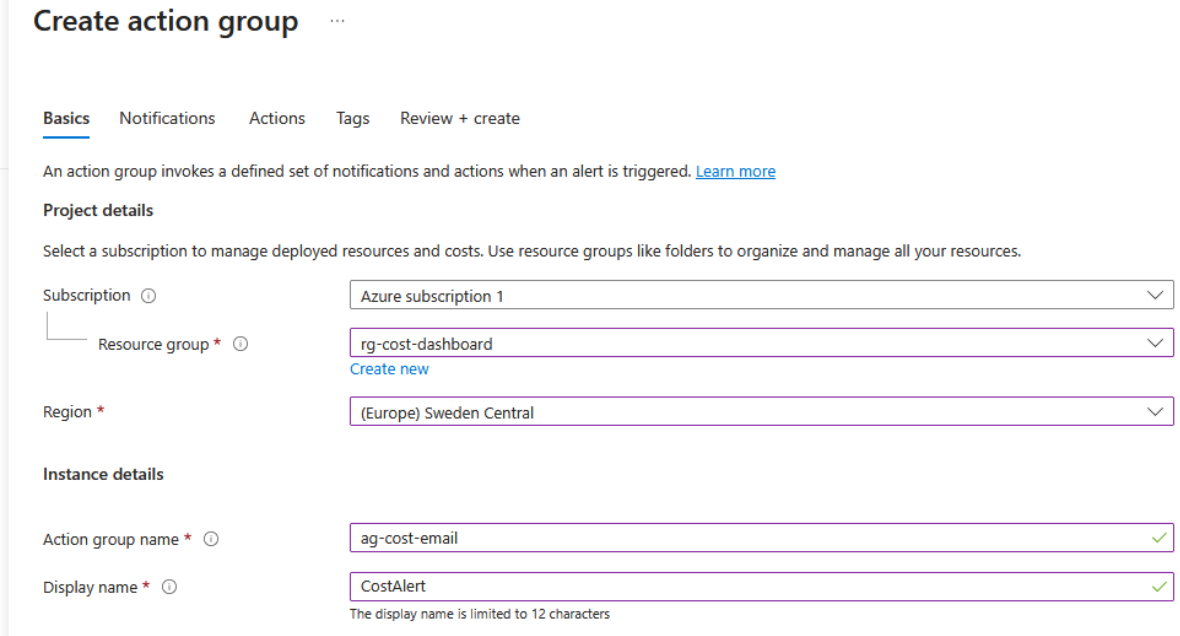


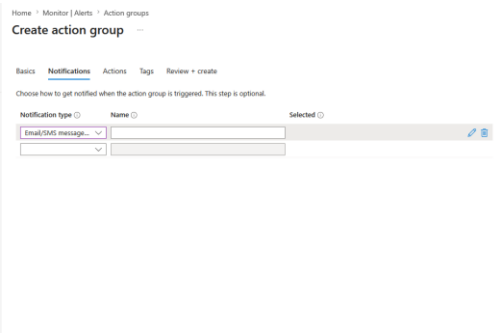
After saving the budget, Azure Cost Management immediately started monitoring the subscription. At this point, the budget exists but no alerts are triggered yet because there is still no resource usage.

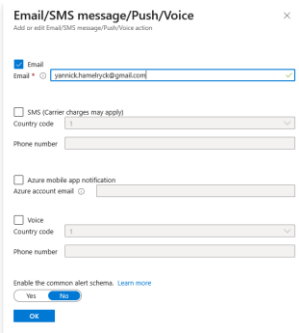


To send notifications automatically, I created an Azure Monitor Action Group. This Action Group is configured to send an email whenever a threshold is reached.
Because the free Azure subscription has some limitations, I could not directly connect the Action Group to the budget. Instead, I used the built-in email recipient option in the budget itself. The Action Group still demonstrates how the solution would normally work in a production subscription.

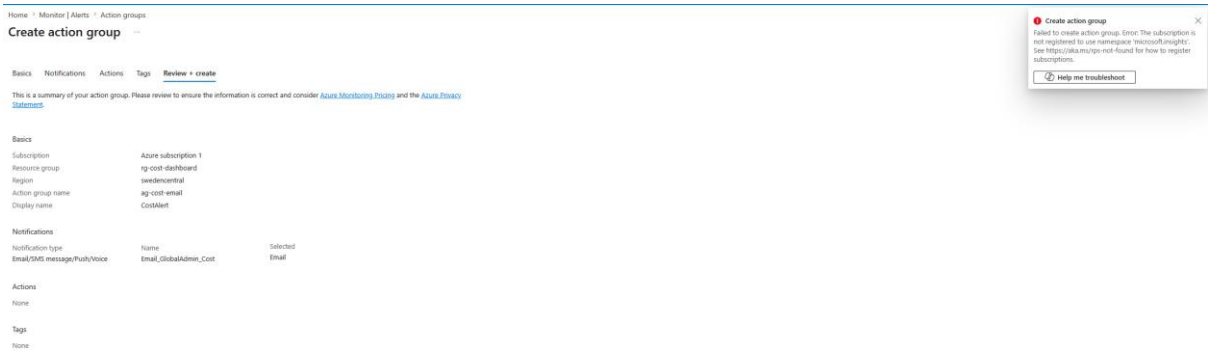


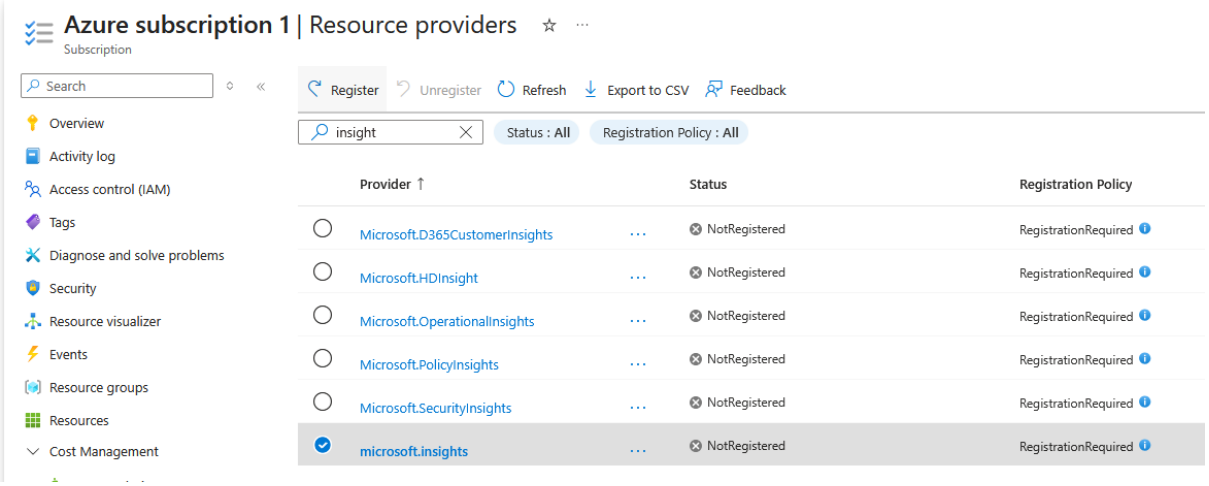


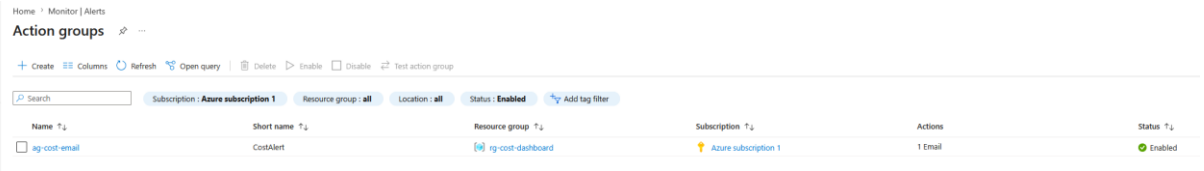


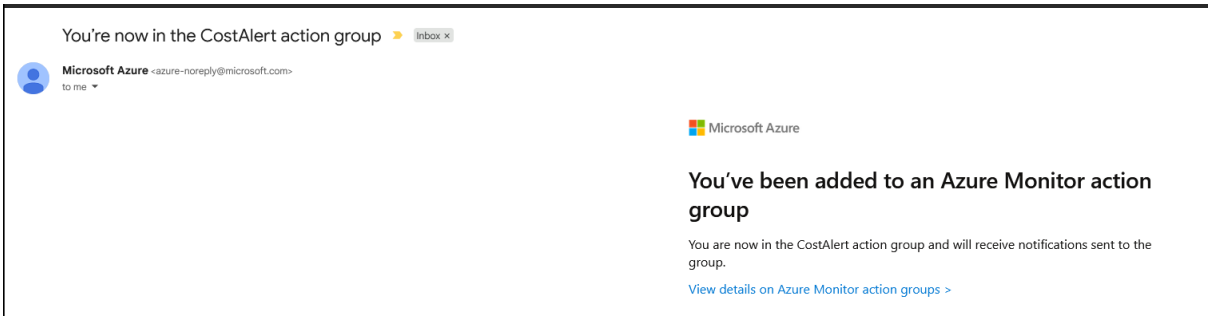


While creating the Action Group, I encountered an error because the “Microsoft.Insights” resource provider was not registered yet in the subscription. After manually registering the required resource providers, the Action Group could be created successfully.

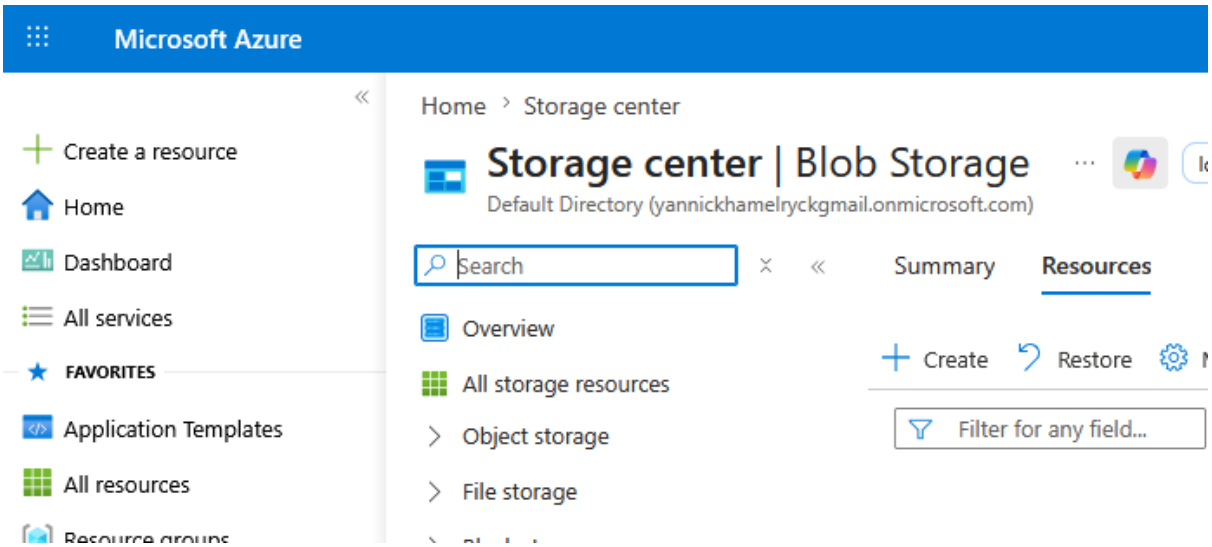








To generate some cost data, I created an Azure Files storage account. I used Standard performance and locally redundant storage (LRS) because it is the cheapest option.
A file share was then created and several files were uploaded. Even a very small amount of storage usage is enough to generate cost information in Azure Cost Management.



Create a storage account ...

Basics | Advanced | Networking | Data protection | Security | Encryption | Tags | Review + create

Azure Storage is a Microsoft-managed service providing cloud storage that is highly available, secure, durable, scalable, and redundant. Azure Storage includes Azure Blobs (objects), Azure Data Lake Storage Gen2, Azure Files, Azure Queues, and Azure Tables. The cost of your storage account depends on the usage and the options you choose below.
Learn more about Azure storage accounts

Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription * : Azure subscription 1

Resource group * : rg-cost-dashboard
Create new

Instance details

Storage account name * : dashboardstorageyannick

Region * : (Europe) West Europe
Deploy to an Azure Extended Zone

Preferred storage type : Azure Files

This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. Learn more

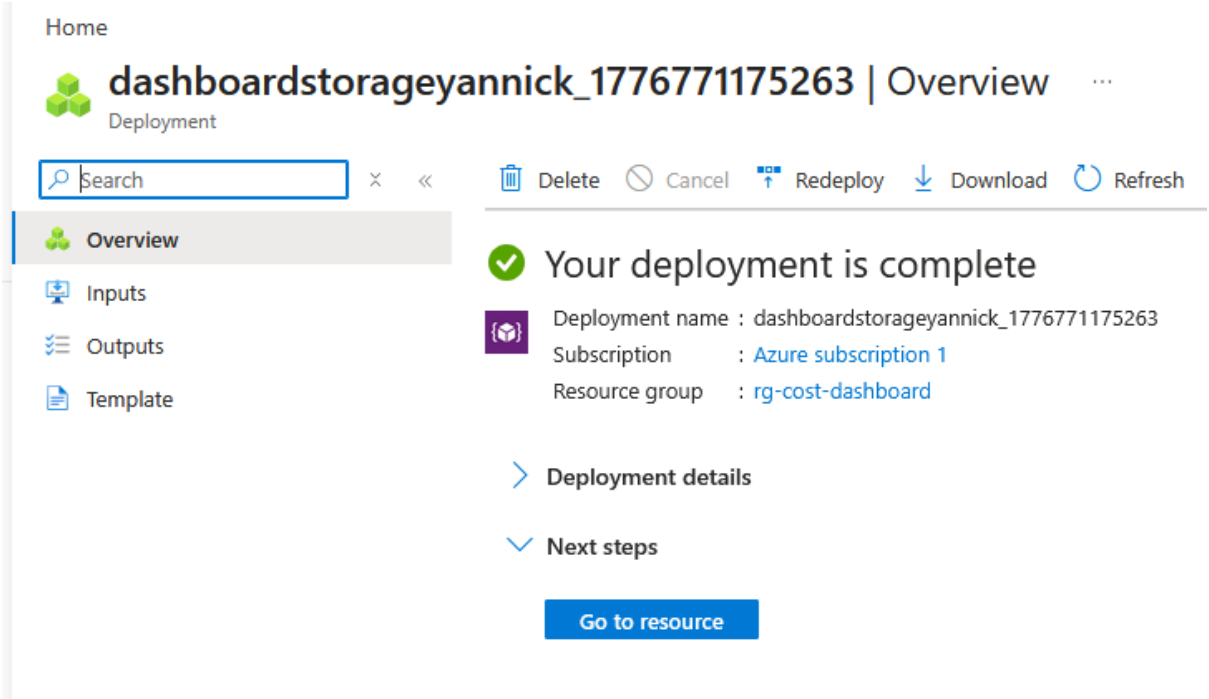
Performance * :

- (●) **Standard:** Recommended for general purpose file shares and cost sensitive applications, such as HDD file shares
- ( ) **Premium:** Recommended for application requiring low-latency or high IOPS/ throughput, such as SSD file shares

File share billing :

- ( ) **Pay-as-you-go file shares:** Billed based on usage
- (●) **Provisioned v2:** Provision capacity, throughput, and IOPS individually (new)

Redundancy * : Locally-redundant storage (LRS)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dashboardstorageyannick_1776771175263 | Overview
Deployment
Search
Overview
Inputs
Outputs
Template
Delete
Cancel
Redeploy
Download
Refresh
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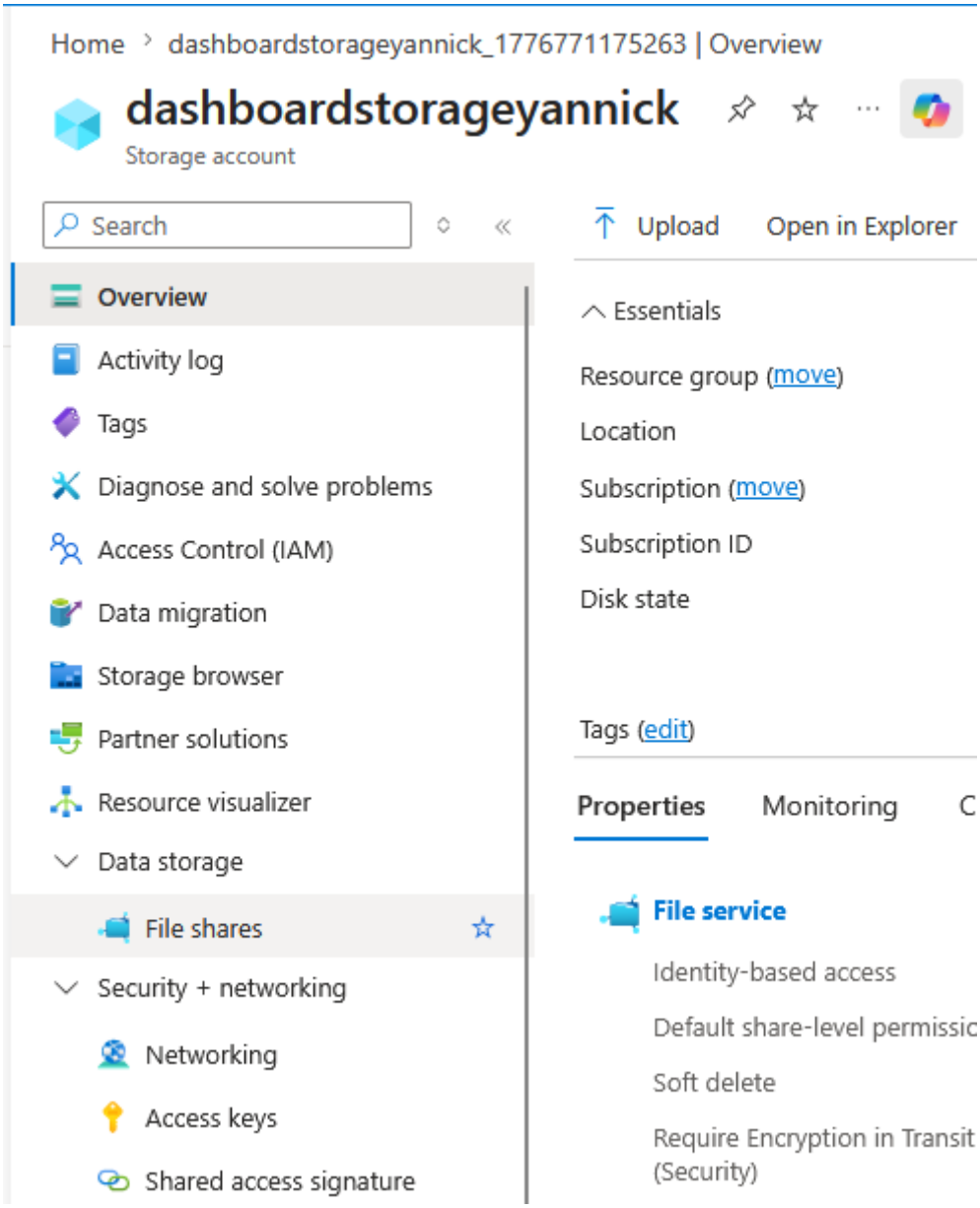
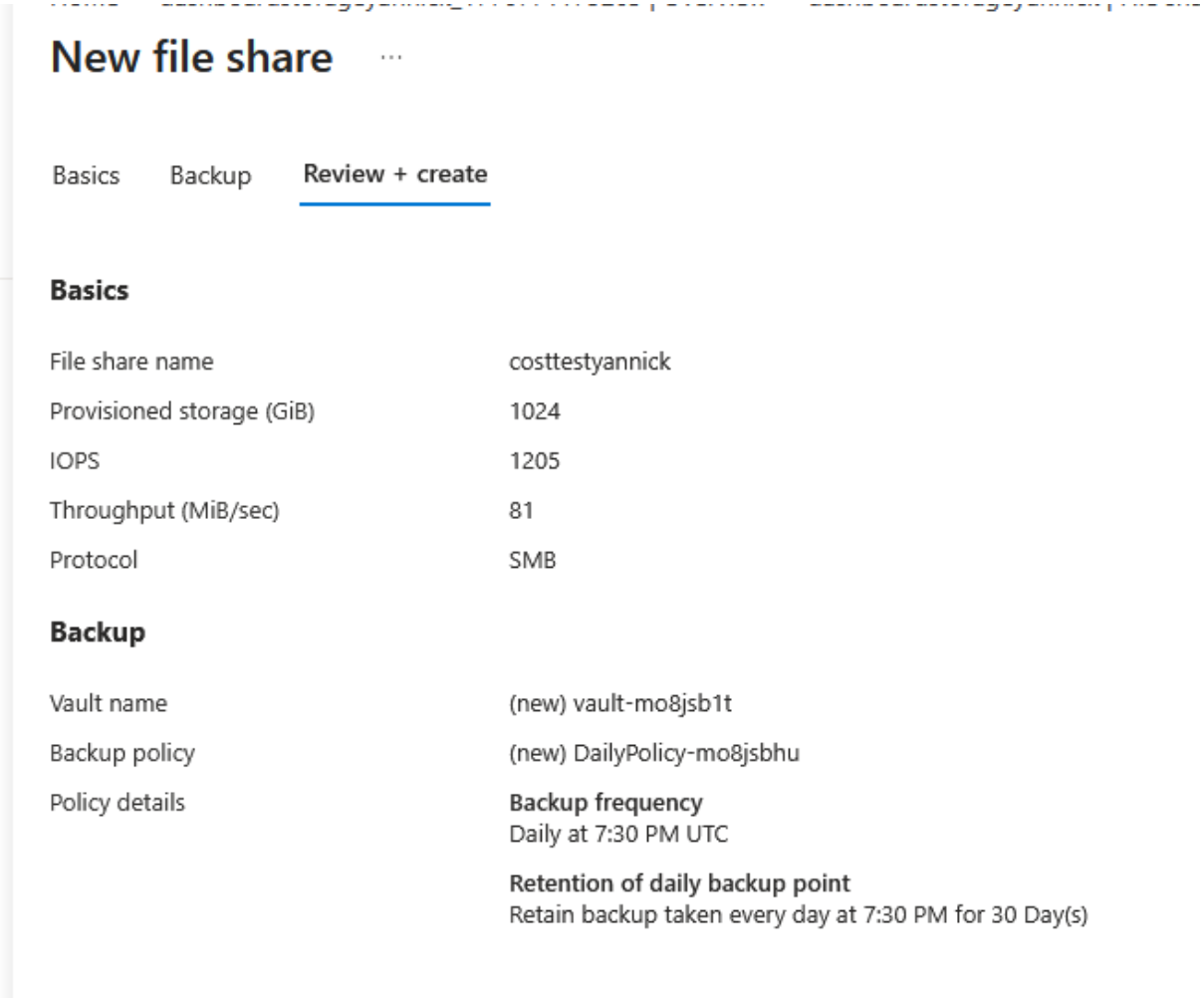
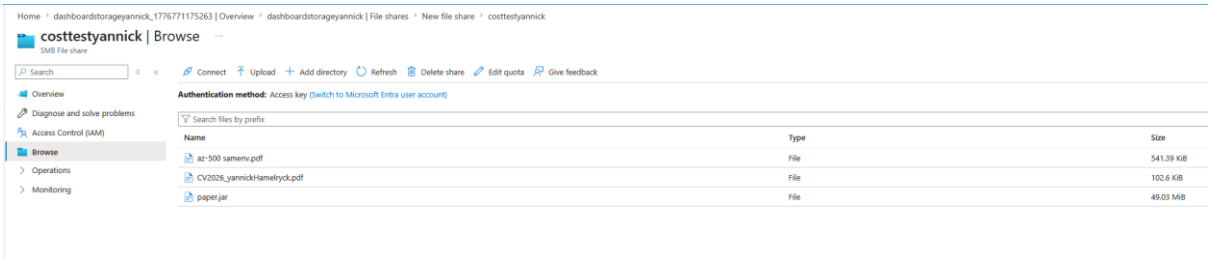
Subscription : Azure subscription 1
Resource group : rg-cost-dashboard
Deployment details
Next steps
Go to resource

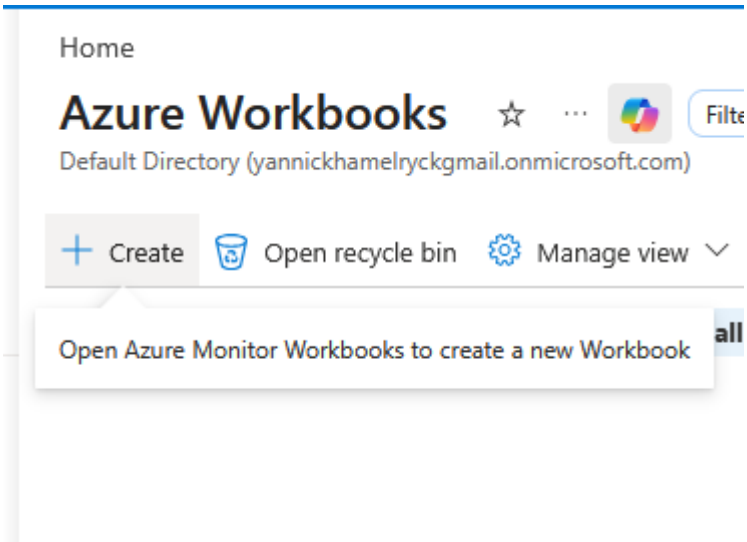

Home
dashboardstorageyannick_1776771175263 | Overview
dashboardstorageyannick
Storage account
Search
Upload
Open in Explorer
Overview
Activity log
Tags
Diagnose and solve problems
Access Control (IAM)
Data migration
Storage browser
Partner solutions
Resource visualizer
Data storage
File shares
Security + networking
Networking
Access keys
Shared access signature
Essentials
Resource group (move)
Location
Subscription (move)
Subscription ID
Disk state
Tags (edit)
Properties
Monitoring
File service
Identity-based access
Default share-level permissic
Soft delete
Require Encryption in Transit (Security)





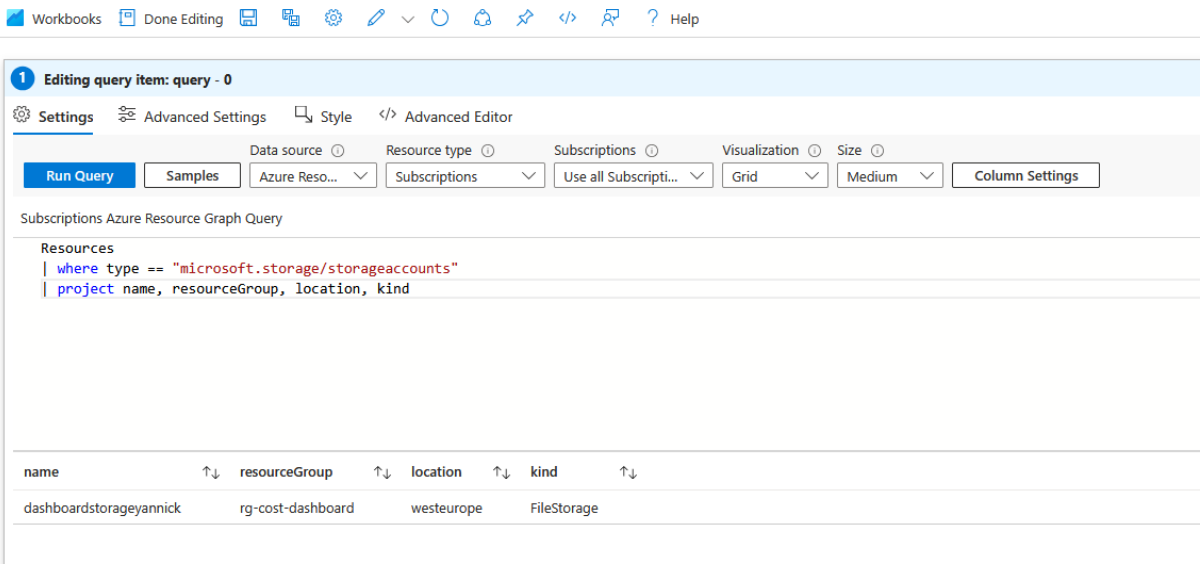
Once the files were uploaded, the project started generating billable usage. Azure Cost Management does not show this information immediately, normally it takes between 8 and 24 hours before the first costs become visible.

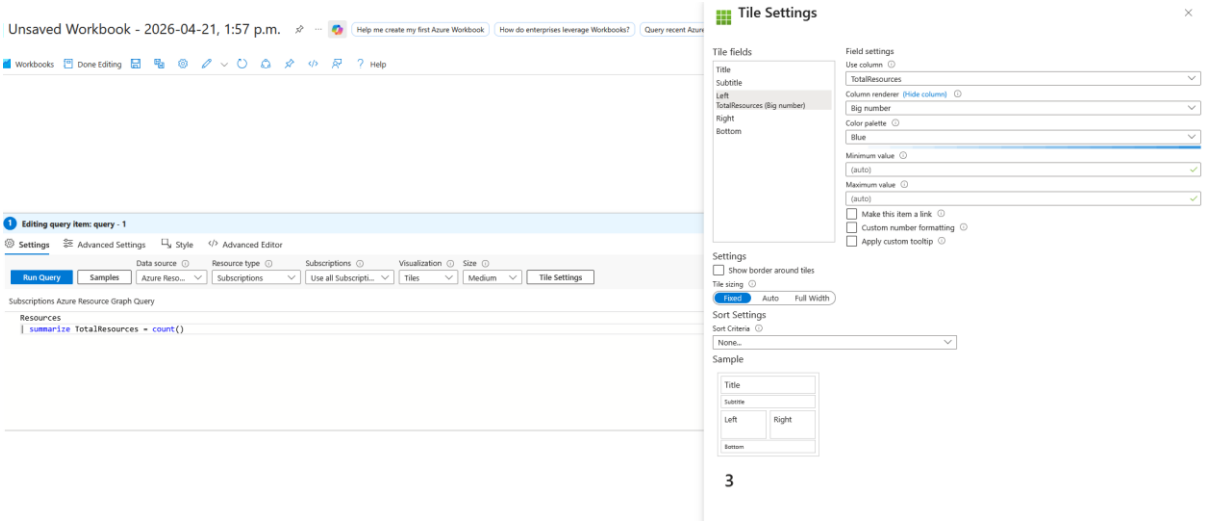
After configuring the budget and generating some usage, I started building the dashboard in Azure Workbooks. The workbook will later visualize the spending and show which Azure resources are responsible for the costs.



Because I already have experience with KQL from working with Microsoft Sentinel, I decided to use Azure Resource Graph queries inside the workbook.

The first query lists all Azure resources inside the project resource group. This was used as a temporary placeholder until the real cost data becomes available.




Unsaved Workbook - 2026-04-21, 1:57 p.m.
Workbooks
Done Editing
Help
Editing query item: query - 1
Settings
Advanced Settings
Style
Advanced Editor
Run Query
Samples
Data source
Resource type
Subscriptions
Visualization
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Resources
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Field settings
Use column
TotalResources
Column renderer (Hide column)
Big number
Color palette
Blue
Minimum value
(auto)
Maximum value
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Make this item a link
Custom number formatting
Apply custom tooltip
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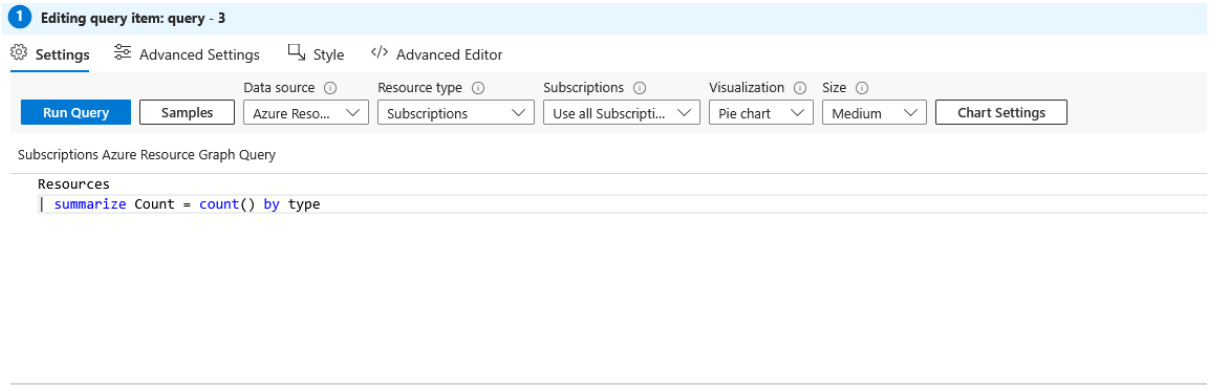
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Settings
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Advanced Editor
Run Query
Samples
Data source
Azure Reso...
Resource type
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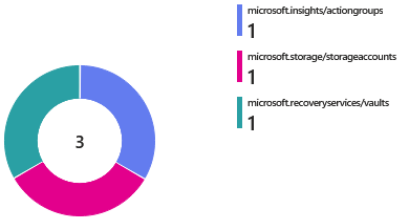
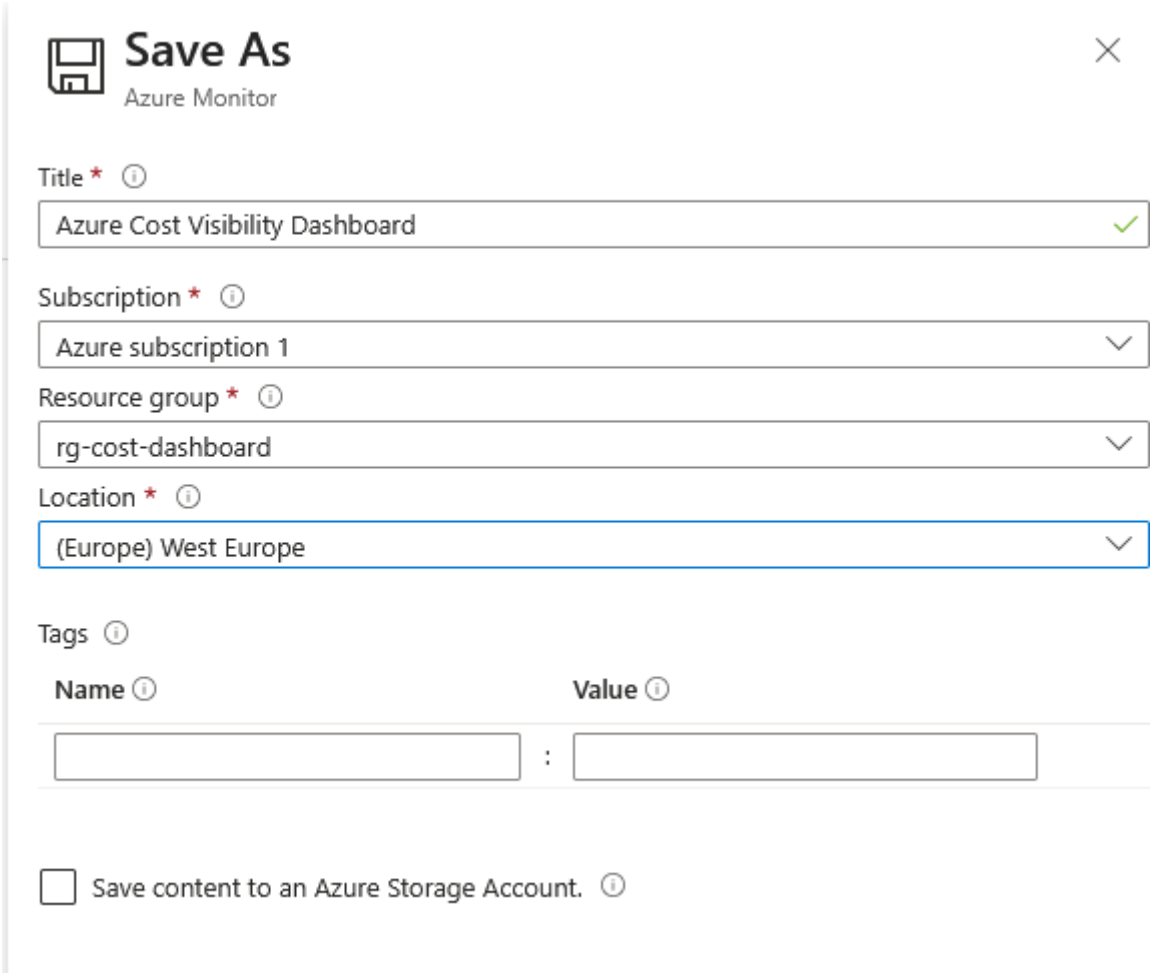
Subscriptions
Use all Subscripti...
Visualization
Pie chart
Size
Medium
Chart Settings
Subscriptions Azure Resource Graph Query
Resources
| summarize Count = count() by type


3
microsoft.insights/actiongroups
1
microsoft.storage/storageaccounts
1
microsoft.recoveryservices/vaults
1


Save As
Azure Monitor
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Subscription *
Azure subscription 1
Resource group *
rg-cost-dashboard
Location *
(Europe) West Europe
Tags
Name
Value
Save content to an Azure Storage Account.

Home > Azure Workbooks > Monitor
Monitor | Workbooks | Azure Cost Visibility Dashboard
Microsoft
How do enterprises leverage Workbooks?
Help me create my first Azure Workbook
Query recent Azure Monitor workbooks changes
Search
Workbooks
Edit
Help
Auto refresh: Off
Overview
Activity log
Alerts
Issues (preview)
Metrics
Logs
Change Analysis
Service health
Workbooks
Dashboards with Grafana
Insights
Managed Services
Settings
Support + Troubleshooting
rg-cost-dashboard
3
microsoft.insights/actiongroups
1
microsoft.storage/storageaccounts
1
microsoft.recoveryservices/vaults
1

[COST DATA not available yet - update tomorrow]

7. WHAT I LEARNED

After completing the project, I successfully created:

- A resource group for the project
- A monthly Azure budget with thresholds
- Email notifications for budget warnings
- An Action Group in Azure Monitor
- A storage account that generates billable usage
- An Azure Workbook dashboard

At the moment of writing, Azure Cost Management still needs additional time before the first cost values became available. The dashboard already works and can later be expanded with real cost metrics.

8. POSSIBLE IMPROVEMENTS [TO DO]

1. Make the dashboard more prettier to look at
2. Use Logic app to send email notifications instead of the default email alert
3. Add Microsoft Teams notifications
4.